

AMI – AMAZON MACHINE IMAGE

❖ What is an AMI?

- An Amazon Machine Image (AMI) is a template that contains the software configuration required to launch an EC2 instance.
- It includes:
 - **Operating system** (e.g., Amazon Linux, Ubuntu, Windows)
 - **Pre-installed software** (web servers, databases, custom applications)
 - **Settings and permissions**
 - **Root volume and data volumes**

When you launch an instance, you select an AMI to define the initial state of the machine

❖ Types of AMIs

- Amazon provides multiple types of AMIs depending on the use case:

Type	Description
AWS-provided AMIs	Official images provided and maintained by AWS (e.g. Amazon Linux, Windows Server)
Marketplace AMIs	Third-party AMIs available for purchase or subscription in AWS Marketplace
Community AMIs	AMIs shared publicly by AWS users (may not be verified or supported)
Custom AMIs	User-created AMIs built from an existing EC2 instance

❖ What is a Launch Template?

- A Launch Template in Amazon EC2 is a reusable configuration for launching EC2 instances.
- It **defines** the instance settings such as
 - **AMI ID**
 - **instance type**
 - **key pair**
 - **security groups**
 - **storage configuration**
 - **tags, and network interfaces**

❖ Why Use Launch Templates?

- Launch Templates in AWS simplify and standardize the process of launching EC2 instances by saving all the configuration details in one reusable template.
- Key Benefits:
 - Reusability
 - Version Control
 - Integration with Other AWS Services
 - Parameter Consistency
 - Overrides Supported
 - Automation Friendly

❖ **How do you migrate an EC2 instance from one region to another?**

1. Create an Amazon Machine Image (AMI) of the existing instance.
2. Copy the AMI to the target region.
3. Launch a new EC2 instance from the copied AMI in the target region

❖ **EC2 Instance Purchasing Options: (Day 5 on LMS see)**

Option	Description	Best For
On-Demand	Pay per hour/second , no commitment	Short-term , unpredictable workloads
Reserved Instances (RI)	Commit for 1 or 3 years, and get up to 75% discount	Steady or long-term workloads
Spot Instances	Buy unused EC2 at up to 90% discount, can be interrupted anytime	Flexible , fault-tolerant tasks
Savings Plans	Commit to spend (\$/hr) over 1 or 3 years, flexible across instance type	Cost savings with flexibility
Capacity Reservations	Reserve capacity in a specific AZ without commitment	Guaranteed capacity in AZ

Interview One-liners:

- ☒ **On-Demand:** "Best for short-term or testing workloads with no commitment."
- ☒ **Reserved:** "Good for predictable, long-term use with big cost savings."
- ☒ **Spot:** "Cheap option for flexible workloads like batch jobs or data analysis."
- ☒ **Savings Plan:** "Offers flexibility and savings like Reserved Instances, but more flexible."

[GENERAL INTERVIEW QUESTIONS]

Q1. What are EC2 Capacity Reservations ?

- EC2 Capacity Reservation allows you to reserve compute capacity (specific instance type in a specific Availability Zone) without long-term commitment.
- Ensures **capacity is available** when you need it.
 - You pay **On-Demand price** for reserved capacity.
 - Useful for **critical workloads** or disaster recovery setups.

Q2. How do they differ from Capacity Reservation & Reserved Instances

Feature	Capacity Reservations	Reserved Instances (RIs)
Purpose	Reserve capacity in a specific AZ	Save cost with long-term commitment
Pricing	Pay On-Demand rate	Pay upfront or partially, up to 75% discount
Commitment	✗ No time commitment	☒ 1 or 3-year commitment
Capacity Guarantee	☒ Yes (guarantees instance availability)	✗ No (only cost-saving, doesn't guarantee capacity)
Flexibility	Flexibility	More flexible, but doesn't hold capacity

Q3. How EC2 Achieves High Network Throughput and what features help in optimizing network performance.

1. Elastic Network Adapter (ENA)

- ENA supports high-performance networking with up to **100 Gbps** throughput.
- Used for modern instance types (e.g., C5n, M5n, R5n).

2. EFA (Elastic Fabric Adapter)

- Specialized for **high-performance computing (HPC)** and low-latency communication.

3. Enhanced Networking

- Provides higher IOPS, lower latency, and more packets per second (PPS).

4. Placement Groups

- **Cluster Placement Group:** Places instances physically close together to reduce latency and increase throughput.

Q4. What is a Spot Fleet, and how does it help in optimizing costs?

- A **Spot Fleet** is a collection of EC2 Spot Instances and, optionally, On-Demand Instances that AWS launched based on a specified target capacity.
- **It helps** in achieving cost optimization by automatically selecting the lowest-priced Spot Instances from different Availability Zones.

Q5. How do you migrate an EC2 instance from one region to another?

1. **Create an Amazon Machine Image (AMI)** of the existing instance.
2. **Copy the AMI** to the target region.
3. Launch a new EC2 instance from the **copied AMI** in the **target region**.

Q6. How do you prevent data loss in EC2 instances?

- To prevent **data loss** in EC2, use **EBS** for persistent storage, take regular snapshots, store important data in **S3**,
- use AMIs for backups, enable **termination protection**, and monitor your instance health with **CloudWatch**.
- These practices ensure high durability, availability, and quick recovery.

Q7. What are the major differences between AWS EC2 Auto Scaling and AWS Elastic Load Balancer (ELB)?

- **Auto Scaling:** Dynamically adjusts the number of EC2 instances based on traffic.
- **Elastic Load Balancer:** Distributes incoming traffic among multiple EC2 instances.

Q8. Explain the concept of EC2 Dedicated Hosts and their use cases.

- An EC2 **Dedicated Host** is a physical server fully dedicated to your use.
- Used for regulatory compliance, **Bring Your Own License (BYOL)** scenarios, and predictable workloads.

Q9. What is the difference between EC2 Hibernate and EC2 Stop?

- **EC2 Stop** shuts down the instance and loses memory state, while
- **Hibernate** saves the RAM and resumes the instance from where it left off. Hibernate is faster to restart **but** uses more EBS storage.